

Molecular Plant

An end-to-end agent-based workflow for genome-reference-independent single-cell annotation in non-model plants --Manuscript Draft--

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Abstract:	We submitted this manuscript as a Correspondence. Here is Abstract of our manuscript. The widespread application of single-cell RNA sequencing (scRNA-seq) in non-model plants is heavily constrained by the absence of high-quality reference genomes and the complexity of pervasive plant paralogs. To address this bottleneck, we developed an end-to-end, multi-agent AI-assisted workflow for genome-reference-independent single-cell annotation. The framework automatically adapts its assembly strategy to NGS, TGS, or hybrid transcriptome inputs to generate custom alignment references, enabling seamless single-cell quantification. For accurate cross-species cell-type knowledge transfer, the pipeline utilizes a refined ortholog mapping strategy that integrates orthologous relationships, phylogenetic information, and sequence identity. Subsequent cell-type interpretation is fully automated via label transfer from model species and LLM-guided evaluation of orthology and functional markers. Validation on a <i>Paeonia lactiflora</i> root dataset confirms that this autonomous system reliably processes raw reads to finalized cell-type labels without expert bioinformatic intervention. By converting intricate computational tasks into natural language interaction, this multi-agent framework effectively eliminates the technical barriers of scRNA-seq analysis, enabling plant biologists to prioritize scientific inquiry over bioinformatics skill acquisition.
Suggested Reviewers:	
Opposed Reviewers:	

Dear Editor,

We are writing to inquire whether our manuscript, tentatively entitled “An end-to-end agent-based workflow for genome-reference-independent single-cell annotation in non-model plants” might be of interest to *Molecular Plant* as a Correspondence.

Single-cell RNA sequencing is increasingly being applied to non-model plants, yet robust cell-type annotation remains difficult in species lacking high-quality reference genomes. In plants, this problem is further complicated by widespread gene duplication and paralog retention, which often make cross-species marker transfer unreliable when based on sequence similarity alone.

To address this, we developed an end-to-end, plant-focused, agent-based workflow implemented within PantheonOS (<https://pantheonos.stanford.edu>). The workflow is designed for genome-reference-independent single-cell annotation in non-model plants and includes three main components: (1) transcriptome-based reference construction and single-cell quantification; (2) an orthology-aware marker transfer strategy based on Best Hit Homologs; and (3) automated cluster consolidation and annotation adjudication.

As a proof of concept, we applied the workflow to a *Paeonia lactiflora* root single-cell dataset. Our results show that a biologically complex non-model plant dataset can be processed from raw sequencing reads to interpretable cell-type labels within a unified workflow, without requiring expert intervention as a necessary step in the core analysis loop.

We would like to be candid that the current manuscript is positioned as a focused methodological Correspondence with proof-of-concept validation, rather than a comprehensive benchmarking study. The present validation is intentionally limited in scale and is centered on one non-model plant dataset together with a modest benchmarking component. We believe that the main value of the study lies in the conceptual and practical advance of a plant-focused, annotation-centered workflow, rather than in broad cross-dataset benchmarking.

We would therefore be very grateful for your advice on whether a manuscript of this scope might be of interest as a Correspondence, or possibly another format, in *Molecular Plant*. If you feel that substantially broader benchmarking or additional validation would be necessary for consideration, we would sincerely appreciate that guidance before formal submission.

We have attached the manuscript draft for your review. Thank you very much for your time and consideration.

Sincerely,

Jie Hao

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An end-to-end agent-based workflow for genome-reference-independent single-cell annotation in non-model plants

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Keywords: non-model plants, single-cell RNA sequencing, ortholog mapping, cell-type annotation, multi-agent framework

Dear Editor,

Single-cell RNA sequencing is increasingly applied to non-model plant species, yet robust cell annotation without a reliable reference genome remains a fundamental challenge. Although transcriptome-based strategies can partially bypass the dependence on high-quality reference genomes, they do not by themselves provide a stable route to cell type annotation^{1,2}. The remaining difficulties can be broadly attributed to three interconnected issues. First, cross-species marker transfer is often ambiguous due to large genome size, high repeat content, and pervasive paralogs³. Second, analytical steps tend to be disconnected; rather than being refined iteratively, clustering results, marker gene evidence, ortholog relationships, and feature-level patterns are typically evaluated in isolation⁴. Third, reproducibility is limited because annotation often relies on manual, experience-driven judgments that are difficult to formalize or trace⁵. Together, these challenges suggest that the core bottleneck is not just data limitations, but the absence of a coherent analytical framework capable of context-aware reasoning and transparent decision-making system.

Recent progress in intelligent workflow systems provides a practical route to reducing this burden. Across computational biology, workflows are increasingly moving from static assistance toward coordinated systems that can execute multi-step analyses and connect intermediate results across a pipeline. Yet automation alone is not enough for the plant cell annotation problem^{6,7}. Generic systems are not designed to handle paralog-rich genomes, uncertain cross-species marker transfer, or the need to refine clusters and adjudicate labels in a biologically coherent way⁸. For non-model plants, meaningful annotation requires more than task orchestration; it requires decision rules that take plant genome complexity into account.

Here, we present an end-to-end agent-assisted workflow designed for genome-reference-independent single-cell annotation in non-model plants. The DPSCA (De novo Plant Single-Cell Annotator) workflow was built on PantheonOS⁹, but its design was driven by the annotation problem itself. It integrates transcriptome reference construction, single-cell quantification, ortholog inference, automated cluster consolidation, and final cell type annotation within a single analysis path. To establish reference transcriptomes across bulk RNA-seq, we implemented a multi-strategy automated assembly framework. This pipeline supports de novo assembly of short reads (NGS) by Trinity, consensus transcript extraction from PacBio long reads by Iso-Seq3, and hybrid error correction of PacBio reads

utilizing paired NGS data via LorDEC. Following assembly, the transcriptome served as the foundation for constructing a custom single-cell reference using TransDecoder and CellRanger. Subsequently, the CellRanger pipeline was employed to map the reads and generate the single-cell expression count matrix. To interpret this matrix and assign biological identities to the cells, accurate ortholog mapping serves as a central methodological step. Instead of relying solely on BLAST-based sequence similarity tools, our automated ortholog mapping strategy integrates orthogroup (OG) inference, phylogenetic information, and sequence identity. Specifically, OrthoFinder was employed to identify OGs and construct phylogenetic gene trees between the target and model species, while BLASTP was used to evaluate sequence identity. We define the best-matching ortholog pair if the query and model-species genes share the same OG and meet one of three conditions: the closest phylogenetic relationship, the highest sequence identity, or representation as the only target-species gene within the OG. These best-matching pairs are more likely to retain annotation relevance than dispersed homologs, thereby reducing false-positive marker transfer. By treating ortholog mapping as the anchor of subsequent annotation, rather than as a preliminary convenience, the workflow is designed to preserve annotation reliability in duplication-rich plant genomes.

To identify an appropriate level of cluster consolidation, our pipeline evaluates a resolution gradient (0.5–2.0) and automatically coalesces clusters sharing highly correlated marker profiles ($R > 0.6$, $p \leq 0.05$). Final annotation is then performed on the refined clusters using a dual-method strategy: (1) orthology-based label transfer from model plants (here we used dataset from the PCMDB), and (2) Large Language Model (LLM, GLM-4.7 for this study)¹⁰-assisted interpretation of marker gene functions. The workflow incorporates an automated integration step to synthesize results from both approaches, yielding a unified annotation result. Through this design, the pipeline aims to deliver consistent cell-type annotations while minimizing the need for specialized bioinformatics expertise.

We evaluated the workflow using a single-cell dataset of *Paeonia lactiflora* (peony) root, a biologically relevant proof-of-concept system lacking a high-quality reference genome. Starting from transcriptome and single-cell sequencing reads, the workflow completed transcript-based reference generation, quantification, orthology-aware marker transfer, cluster refinement, and final annotation within one connected pipeline. Following quality control, 12,842 high-quality cells were retained for subsequent analysis and subsequently consolidated into 18 refined clusters. Leveraging the integrated annotation framework, the pipeline assigned biologically interpretable root cell identities, including

phloem, quiescent center, root cortex, root endodermis, trichoblast, and xylem. Notably, rare cell populations, such as the stem cell niche and a portion of meristem tissue, were independently annotated by both orthology mapping and LLM-assisted marker interpretation, providing convergent support for the inferred identities. Although this application is intended as a proof of concept rather than a complete atlas, it demonstrates that a non-model plant dataset can be processed from raw reads to interpretable cell-type labels within a unified, annotation-centered workflow with the minimal expert intervention.

The value of this result lies not only in the peony application itself, but also in what it clarifies about the annotation problem in plants. In genome-reference-independent settings, the principal obstacle is not simply the lack of a complete genome assembly, nor merely the inconvenience of multistep computation. Rather, it is the instability of cross-species annotation when ortholog inference is treated too simplistically in paralog-rich genomic contexts. Once this problem is addressed at the level of marker transfer, subsequent annotation steps become much more tractable: cluster consolidation becomes easier to justify, and annotation adjudication becomes more coherent, because both operate on a more reliable homolog-mapping basis.

More broadly, our results suggest that orthology-aware marker transfer should be regarded as a central requirement, rather than an optional refinement, for automated annotation in plant single-cell analysis. As single-cell studies continue to expand into medicinal plants, woody perennials, crops with complex ploidy histories, and other non-model lineages, the field will increasingly need workflows that do more than automate file handling and tool execution. What is needed is a framework that connects transcript-based quantification to evolution-aware interpretation while reducing, rather than merely relocating, the burden of expert judgment. In this sense, the present workflow provides a practical step toward making non-model plant single-cell datasets more executable, more reproducible, and more tractable for routine analysis across diverse species.

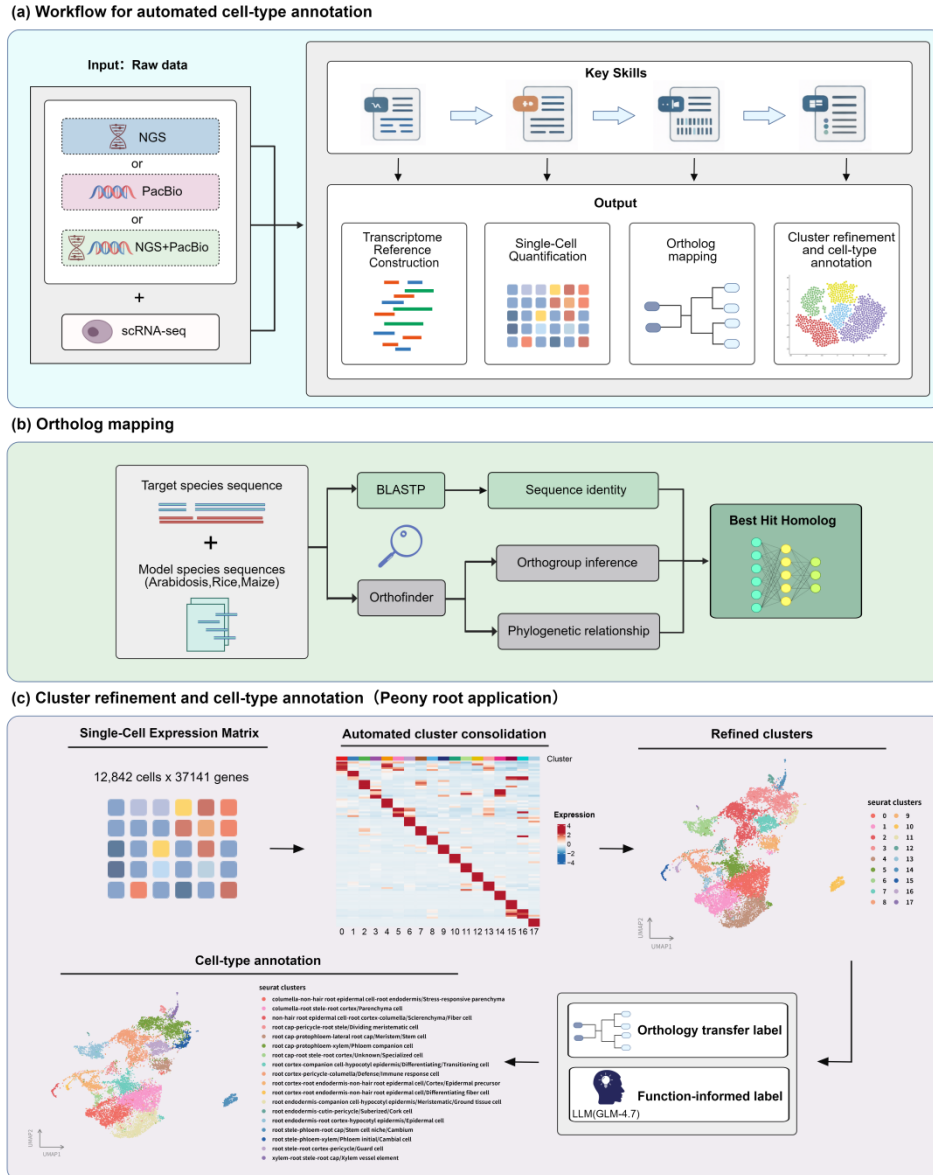


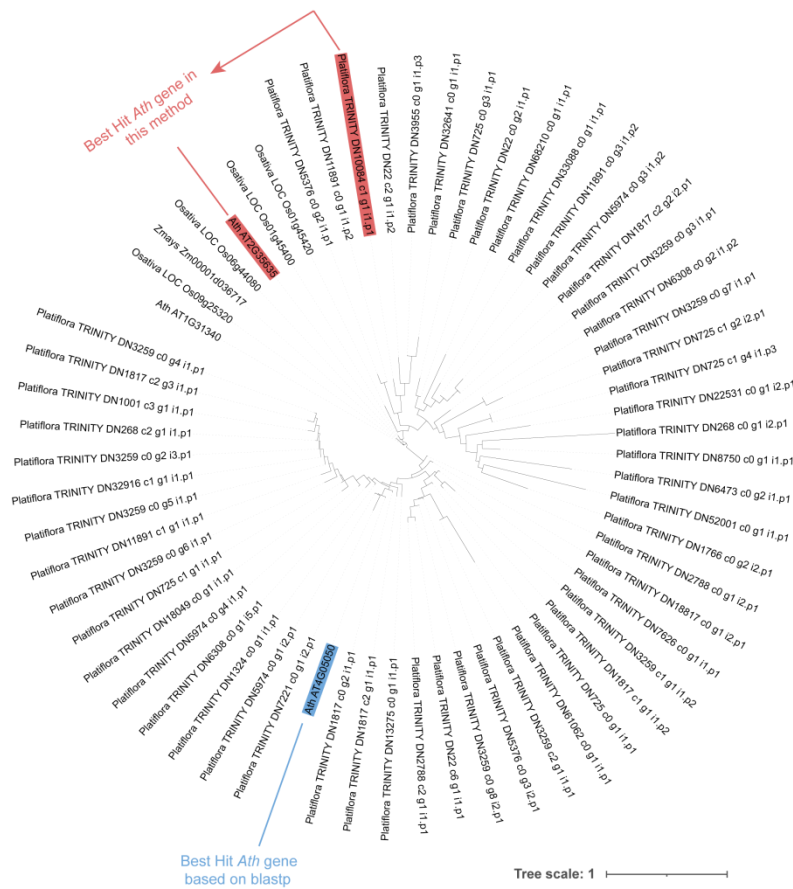
Figure 1: Automated cell type annotation workflow for non-model plant single-cell data.

(a) Workflow overview. Raw sequencing data, including NGS, PacBio, or combined NGS+PacBio and scRNA-seq inputs, are used for transcriptome reference construction, single-cell quantification, ortholog mapping, cluster refinement, and cell-type annotation. (b) Ortholog mapping module. Target-species sequences are evaluated against model species reference by integrating sequence identity, orthogroup inference, and phylogenetic relationship to derive informative homologs for annotation. (c) Peony root application. The single-cell expression matrix is first subjected to automated cluster consolidation to obtain refined clusters, followed by cell-type annotation through integration of homolog transfer label and function-informed label. This workflow is designed to support reliable cell-type assignment and does not extend to post-annotation downstream analysis.

References

1. Xue, H.-C., Xu, Z.-G., Liu, Y.-J., Wang, L., Ming, X., Wu, Z.-Y., Lian, H., Han, Y.-W., Xu, J., Zhang, Z.-D., et al. (2025). A unified cell atlas of vascular plants reveals cell-type foundational genes and accelerates gene discovery. *Cell* *0*. <https://doi.org/10.1016/j.cell.2025.07.036>.
2. Grones, C., Eekhout, T., Shi, D., Neumann, M., Berg, L.S., Ke, Y., Shahan, R., Cox, K.L., Gomez-Cano, F., Nelissen, H., et al. (2024). Best practices for the execution, analysis, and data storage of plant single-cell/nucleus transcriptomics. *Plant Cell* *36*, 812–828. <https://doi.org/10.1093/plcell/koac003>.
3. Passalacqua, M.J., and Gillis, J. (2024). Coexpression enhances cross-species integration of single-cell RNA sequencing across diverse plant species. *Nat. Plants* *10*, 1075–1080. <https://doi.org/10.1038/s41477-024-01738-4>.
4. Zhang, Y., Sun, H., Zhang, W., Fu, T., Huang, S., Mou, M., Zhang, J., Gao, J., Ge, Y., Yang, Q., et al. (2024). CellSTAR: a comprehensive resource for single-cell transcriptomic annotation. *Nucleic Acids Res.* *52*, D859–D870. <https://doi.org/10.1093/nar/gkad874>.
5. Chen, H., Liang, P., Lu, X., Jiang, S., Jin, J., Cai, J., Lu, Y., Zheng, D., Yao, J., Chu, Q., et al. (2026). The plant ontology of cell types. *Mol. Plant* *19*, 425–429. <https://doi.org/10.1016/j.molp.2025.12.025>.
6. Alber, S., Chen, B., Sun, E., Isakova, A., Wilk, A.J., and Zou, J. (2026). CellVoyager: AI CompBio Agent Generates New Insights by Autonomously Analyzing Biological Data. *Nat. Methods*. <https://doi.org/10.1038/s41592-026-03029-6>.
7. Huang, K., Zhang, S., Wang, H., Qu, Y., Lu, Y., Li, R., Qiu, L., Li, G., Zhang, J., Yin, D., et al. (2025). Biomni: A General-Purpose Biomedical AI Agent. Preprint at bioRxiv. <https://doi.org/10.1101/2025.05.30.656746>.
8. Chau, T.N., Timilsena, P.R., Bathala, S.P., Kundu, S., Bargmann, B.O.R., and Li, S. (2025). Orthologous marker groups reveal broad cell identity conservation across plant single-cell transcriptomes. *Nat. Commun.* *16*, 201. <https://doi.org/10.1038/s41467-024-55755-0>.
9. Xu, W., Poussi, E., Zhong, Q., Zeng, Z., Zou, C., Wang, X., Lu, Y., Cui, M., Okamura, D., Huang, C., et al. (2026). PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery. Preprint at bioRxiv. <https://doi.org/10.64898/2026.02.26.707870>.
10. GLM, T., Zeng, A., Xu, B., Wang, B., Zhang, C., Yin, D., Zhang, D., Rojas, D., Feng, G., Zhao, H., et al. (2024). ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools. Preprint at arXiv, <https://doi.org/10.48550/arXiv.2406.12793>.

Supplementary file



Supplementary Figure 1: Schematic illustration of identifying the optimal ortholog mapping among many-to-many homologous genes. The *Arabidopsis* gene highlighted in red is the ortholog we identified for the *P. lactiflora* gene TRINITY_DN10084_ci_g1_i1.p1. The gene highlighted in blue represents the *Arabidopsis* homolog selected based on the BLASTP score.

Data and code availability

The raw sequencing reads for *Paeonia lactiflora* root are available from the corresponding author on request. The skill files of DPSCA and example results are available on the GitHub website (<https://github.com/Hao-Zou-lab/DPSCA>).

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Acknowledgments

The authors have no conflicts to declare.

Author contributions

J.H., X.Z., X.Y., Z.X., L.G., and H. T. designed this study. X.Y., Z.X., Y.T., and X.M. performed bioinformatic analysis, X.Y., Z.X., Y.T., and Y.M. wrote the manuscript; J.H. and X.Z. revised the manuscript; L.W., and S.Y. provided help in the plant and RNA sample preparation; and X.Y., Y.T., and J.W. prepared the figures.; All authors read and approved the final manuscript.

Supplementary Figure 1: Schematic illustration of identifying the optimal ortholog mapping among many-to-many homologous genes. The *Arabidopsis* gene highlighted in red is the ortholog we identified for the *P. lactiflora* gene TRINITY_DN10084_ci_g1_i1.p1. The gene highlighted in blue represents the *Arabidopsis* homolog selected based on the BLASTP score.

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